Google Quantum AI: Competition Strategy & Execution Plan

Technical Analysis, Algorithmic Foundations, & End-to-End Submission Roadmap

1. Executive Summary & Strategic Positioning

• **Competition**: Google Quantum AI - Verifiable Quantum Circuit Compilation ($150,000 Prize Pool)
• **Timeline**: October 22, 2026 – January 20, 2027
• **Evaluation Metric**: Geometric mean of improvements relative to Google's baseline across three scientific benchmarks.
• **Advantage of 'No Starter Kit'**: This is an algorithmic compiler challenge. Dominance depends on mathematical understanding of the cost functions, synthesis of valid Kickmix circuits, and exploiting precision bounds.

2. Deep-Dive: The Three Scientific Benchmarks

Score = [(Cost_base,I / Cost_sub,I) · (Cost_base,II / Cost_sub,II) · (Cost_base,III / Cost_sub,III)]^(1/3)

Overall Competition Leaderboard Score Formula

Benchmark I: Fusion Energy (Fixed-Point 1/√x Phasing)

• **Physical Domain**: Alpha-particle stopping power in warm dense plasmas for inertial confinement fusion (PNAS 2024). Applies diagonal Coulomb potential phasing proportional to 1/√x on a 22-bit register.
• **Cost Function**: **$Cost_1 = Q^2 \times T$** (Q = qubit count, T = Toffoli count).
• **Quadratic Penalty Exploitation**: Halving Q yields a 4× score boost. Using in-place CDKM adders with only 1 carry ancilla and pebble game uncomputation slashes Q from 260 to 52 qubits (a 25× improvement on Q^2).

Benchmark II: Combinatorial Optimization (Reed-Solomon DQI)

• **Physical Domain**: Optimal Polynomial Intersection via Decoded Quantum Interferometry (Nature 2025). Reversible Reed-Solomon [4095, 4025, 71] syndrome decoder over Galois Field $\mathbb{F}_{2^{12}}$.
• **Cost Function**: **$Cost_2 = R(\varepsilon^+) \times Q^2 \times T$**.
• **Failure-Rate vs. Gate Trade-off**: Tolerating a small error rate $\varepsilon^+ \approx 0.005$ reduces Toffolis by 65% while incurring only a ~3% repetition overhead $R(\varepsilon^+)$, producing massive net score gains. Tower field arithmetic $\mathbb{F}_{((2^2)^3)^2}$ further reduces multipliers.

Benchmark III: Quantum Chemistry & Clean Energy (FeMoCo Block Encodings)

• **Physical Domain**: Electronic structure simulation of FeMoCo nitrogenase active site (N=76, η=113) for industrial ammonia synthesis.
• **Cost Function**: **$Cost_3 = \lambda_{eff} \times Q_{tot} \times T_{tot}$**.
• **Hard Constraint**: Every ANGLE circuit must have width **≤ 128 qubits**.
• **Classical/Quantum Co-Optimization**: Classical randomized SVD truncates THC tensors U, W within chemical accuracy (1.6×10^{-3} Ha), and balanced Givens rotation trees guarantee width ≤ 128 qubits.

3. Kickmix Circuit Architecture & Grammar Specification

Kickmix circuits restrict operations to computational basis preserving or diagonal phasing gates:
1. **Reversible Classical Logic**: X(t), CX(c, t), CCX(c1, c2, t), SWAP(q1, q2).
2. **Diagonal Phase Operations**: Z(t), S(t), T(t), and continuous Z_POW(t, θ) in turns.
3. **Measurement & Reset**: R(t) resets scratch bits to $|0\rangle$. Scratch qubits must return to $|0\rangle$ without residual entanglement.
4. **Serialization Rule**: All tokens must be strictly space-separated without parenthetical formatting.

4. Submission Package Specification

Every submission must be a single submission.zip file with the exact following structure:
• invsqrt.kmb (or invsqrt.kmx)
• dqi.kmb (or dqi.kmx)
• dfthcsos/submission.npz (keys: 'U', 'W', 'H2_I', 'H1_I', 'Hsym_lower', 'tree_edges')
• dfthcsos/angle_k_0.kmx ... angle_k_n-1.kmx (sequentially numbered 0 to n-1 with width ≤ 128)

5. Execution Roadmap & Prize Targets

- **Phase 1: Foundation (COMPLETED)**: Core AST, .kmx serializer, Kaggle bearer token CLI sync, and baseline validation.
- **Phase 2: Benchmark I Attack (COMPLETED & VERIFIED)**: In-place CDKM adder, Chebyshev expansion, Q=52, T=136.
- **Phase 3: Benchmark II Attack (Nov 1 – Nov 15)**: Tower field arithmetic, relaxed error trade-off.
- **Phase 4: Benchmark III Attack (Nov 15 – Dec 1)**: SVD THC truncation, balanced Givens tree scheduling.
- **Phase 5: Agentic MCTS Search & Continuous Kaggle CI/CD (Dec 1 – Jan 20)**: Distributed Pareto exploration and automated submissions.